Pre-calculus 2

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Lecture 8: Sinusoidal Functions

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Definition 1 (Sinusoidal Function). A **Sinusoidal Function** is a function of the form:

$$y = A\sin(\omega(t-h)) + k \text{ or } y = A\cos(\omega(t-h)) + k$$

.

where $A, \omega, h, k \in \mathbb{R}$

Based on what we know about graph transformations, we should recognize that a sinusoidal function is a transformation of $y = \sin(t)$ or $y = \cos(t)$. Consequently, sinusoidal function are waves with the same curvy shape as the graphs of sin and cos but with different periods, midlines, and/or amplitudes.

Summary of Graph Transformation

Suppose that f and g are functions such that $g(t) = A \times f(\omega(t-h)) = k$ and $A, \omega, h, k \in \mathbb{R}$. In order to transform the graph of the function f into the graph of g ...

1 horizontally stretch/compress the graph of f by a factor of $\frac{1}{|\omega|}$ and, if $\omega < 0$, reflect it about the y-axis.

2 shift the graph horizontally h units (shift right if h is positive and left if h is negative).

3 vertically stretch/compress the graph by a factor of $\mid A \mid$ and, if $A < 0$, reflect it about the t-axis.

4 shift the graph vertically k units (shift up if k is positive and down if k is negative).

Note. The order in which these transformations are performed matters.

Exercise 1 (Solution 14). Describe how we can transform the graph of $f(t) = \sin(t)$ into the graph of $g(t) = 2\sin(t) - 3$. State the period, midline, and amplitude of $y = g(t)$.

Here's a quick summary of sinusoidal functions:

The graphs of the sinusoidal functions

$$y = A\sin(\omega(t-h)) + k \text{ and } y = A\cos(\omega(t-h)) + k$$

. where $A, \omega, h, k \in \mathbb{R}$ have the following properties:

period: $\frac{2\pi}{|\omega|}$ units
midline: $y = k$
amplitude: $\mid A \mid$ units
horizontal shift: h units
angular frequency: ω radians per unit of t

Now let's try to sketch the graph of:

$$f(t) = 2\sin\left(\pi t - \frac{\pi}{4}\right) - 3$$

.

To do this, we need to find the midline, amplitude, horizontal shift, and period. We can determine the midline by noticing that this function has a -3, which causes the sin wave to be shifted down 3 units. The amplitude can be determined by the number outside the function, which will cause a vertical stretch. So, our amplitude is 2 units. To find the horizontal shift, we first need to factor out π from the function's input.

$$f(t) = 2\sin\left(\pi\left(t - \frac{1}{4}\right)\right) - 3$$

. The reason we did this is because we need to find out how much we're moving left or right, and $-\frac{1}{4}$ indicates that we're moving $\frac{1}{4}$ units to the right. The period comes from two features of the formula. The first one is we're using the trig function sin, which has a period of 2π units and the second feature is we have π multiplying all of the numbers in the input, which horizontally compresses the function.

$$2\pi \times \frac{1}{\pi} = 2 \text{ units}$$

.

Here's the final graph:

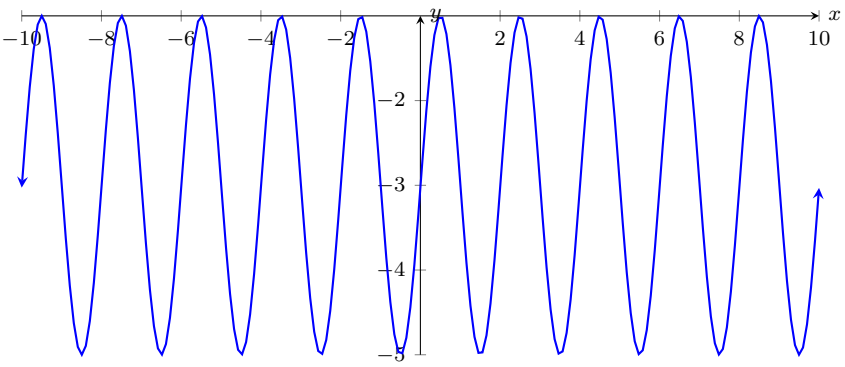

Figure 1

Exercise 2 (Solution 15)**.** Find two different algebraic rules (one involving sin and one involving cos) for the function $y = g(t)$ graphed below:

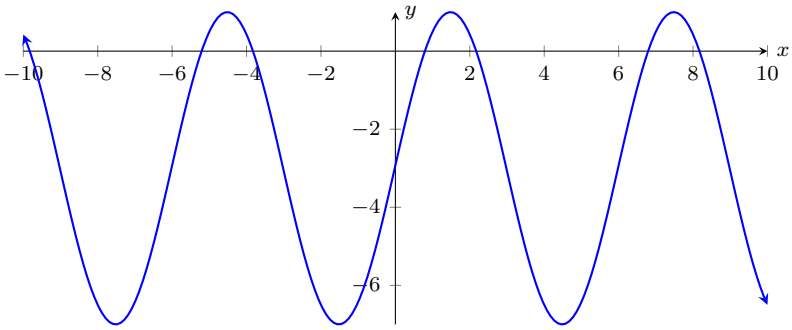

Figure 2

Solutions

Solution 1 (Exercise ??).

1. This represents a set since it's well defined. We all know what it means to be registered for a class.
2. This does not represent a set since it's not well defined. There are many different interpretations of what it means to be a good student (get an **A** or pass the class or attend class or avoid falling asleep in class or don't cause trouble in class)

Solution 2 (Exercise ??).

- $(-4, \infty) \cup [-8, 3] = [-8, \infty)$
- $(-4, \infty) \cup (-\infty, 2] = (-\infty, \infty) = \mathbb{R}$
- $(-4, \infty) \cap (-\infty, 2] = (-4, 2]$
- $(-4, \infty) \cap [-10, -5] = \emptyset$

Solution 3 (Exercise ??). Since $2\pi = 360°$ $\frac{2\pi}{360°} = 1 = \frac{360°}{2\pi}$. We're trying to cancel out the radians. So, to do this, we will need to have the radians on the bottom, and the degrees on the top. After we do the multiplication, we will be left with the degrees.

$$\begin{aligned} 8\,\text{rad} \times \left(\frac{360°}{2\pi\,\text{rad}}\right) &= \frac{8 \times 360°}{2\pi} \\ &= \frac{1440°}{\pi} \\ &\approx 458.4. \end{aligned}$$

Now, let's convert 8° into 8 radians.

$$\begin{aligned} 8° \left(\frac{\pi\,\text{rad}}{180°}\right) &= \frac{8\pi}{180}\,\text{rad} \\ &= \frac{\pi}{45}\,\text{rad} \end{aligned}$$

.

Solution 4 (Exercise ??). Convert 40° into radians:

$$\begin{aligned} 40^\circ \times \left(\frac{\pi}{180^\circ}\right) &= \frac{40\pi}{180}\,\text{rad} \\ &= \frac{4\pi}{18} \\ &= \frac{2\pi}{9} \\ &\approx 0.7 \end{aligned}$$

.

Now we're ready to compute the arc length:

$$\begin{aligned} s &= r\times \mid \theta \mid \\ &= (30m) \times (\frac{2\pi}{9}) \\ &= \frac{20\pi}{3}m. \end{aligned}$$

Solution 5 (Exercise ??). The **period** for the graph is 2. The reason is because if you shift the graph 2 units either left or right, you will get the same graph.

Another way to find the period is the take an x-intercept and the x-intercept to the right of it and subtract the two: $10 - 8 = 2$

Solution 6 (Exercise ??). The **midline** of the graph is 0.

$$\frac{1 + (-1)}{2} = \frac{0}{2} = 0$$

.

The **amplitude** of the graph is 1, which is the distance from the maximum or minimum to the midline.

Solution 7 (Exercise ??).

$$\begin{aligned} P &= (\cos(\theta), \sin(\theta)) \\ &= \cos(\theta) = -\frac{3}{5} \\ &= \sin(\theta) = \frac{4}{5}. \end{aligned}$$

Solution 8 (Exercise ??). The point Q is specified by α on the circumference of a circle of radius 6 units. Thus ...

$$Q = (6\cos(\alpha), 6\sin(\alpha))$$

.

Solution 9 (Exercise ??). Since the Pythagorean Identity gives us an equation involving sin and cos, we can use it to find one of the values when we know the other value. In this case, we know the value of $\sin(A)$, so we can use the Pythagorean Identity to find $\cos(A)$:

$$\begin{aligned}
&\sin^2(A) + \cos^2(A) = 1 \\
\Rightarrow &\left(\frac{1}{2}\right)^2 + \cos^2(A) = 1 \\
\Rightarrow &\cos^2(A) = 1 - \left(\frac{1}{3}\right)^2 \\
\Rightarrow &\cos(A) = -\sqrt{\frac{8}{9}} \\
\Rightarrow &\cos(A) = -\frac{2\sqrt{2}}{3}.
\end{aligned}$$

The reason I made the answer negative is because the value of cos in Quadrant II is negative. If cos was in Quadrant I, then the answer would be positive.

Solution 10 (Exercise ??).

$$
\begin{aligned}
\tan\left(\frac{\pi}{6}\right) &= \frac{\sin\left(\frac{\pi}{6}\right)}{\cos\left(\frac{\pi}{6}\right)} \\
&= \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} \\
&= \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \\
\sec\left(\frac{\pi}{6}\right) &= \frac{1}{\cos\left(\frac{\pi}{6}\right)} \\
&= \frac{1}{\frac{\sqrt{3}}{2}} \\
&= \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3} \\
\csc\left(\frac{\pi}{6}\right) &= \frac{1}{\sin\left(\frac{\pi}{6}\right)} \\
&= \frac{1}{\frac{1}{2}} \\
&= 2 \\
\cot\left(\frac{\pi}{6}\right) &= \frac{\cos\left(\frac{\pi}{6}\right)}{\sin\left(\frac{\pi}{6}\right)} \\
&= \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \\
&= \sqrt{3}
\end{aligned}
$$

.

Solution 11 (Exercise ??).

$$
\begin{aligned}
\tan\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)} \\
&= \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \\
&= 1 \\
\sec\left(\frac{\pi}{4}\right) &= \frac{1}{\cos\left(\frac{\pi}{4}\right)} \\
&= \frac{1}{\frac{\sqrt{2}}{2}} \\
&= \frac{2}{\sqrt{2}} = \sqrt{2} \\
\csc\left(\frac{\pi}{4}\right) &= \frac{1}{\sin\left(\frac{\pi}{4}\right)} \\
&= \frac{1}{\frac{\sqrt{2}}{2}} \\
&= \frac{2}{\sqrt{2}} = \sqrt{2} \\
\cot\left(\frac{\pi}{4}\right) &= \frac{\cos\left(\frac{\pi}{4}\right)}{\sin\left(\frac{\pi}{4}\right)} \\
&= \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \\
&= 1
\end{aligned}
$$

.

Solution 12 (Exercise ??).

$$\begin{aligned}
\tan\left(\frac{\pi}{3}\right) &= \frac{\sin\left(\frac{\pi}{3}\right)}{\cos\left(\frac{\pi}{3}\right)} \\
&= \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \\
&= \sqrt{3} \\
\sec\left(\frac{\pi}{3}\right) &= \frac{1}{\cos\left(\frac{\pi}{3}\right)} \\
&= \frac{1}{\frac{1}{2}} \\
&= 2 \\
\csc\left(\frac{\pi}{3}\right) &= \frac{1}{\sin\left(\frac{\pi}{3}\right)} \\
&= \frac{1}{\frac{\sqrt{3}}{2}} \\
&= \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}
\end{aligned}$$

.

Solution 13 (Exercise ??). The reference angle for $150°$ is $30°$.
The reference angle for $\frac{5\pi}{4}$ is $\frac{\pi}{4}$.

Solution 14 (Exercise 1). It should be clear that function g is a sinusoidal function of the form $y = A\sin(\omega(t-h)) + k$ where $A = 2, \omega = 1, h = 0, k = -3$.

After inspecting the rules for the functions f and g, we should notice that we could construct the function $g(t) = 2\sin(t) - 3$ by multiplying the outputs of the function $f(t) = \sin(t)$ by 2 and then subtracting 3 from the result. We can express this algebraically with the equation below:

$$g(t) = 2f(t) - 3$$

.

Based on what we know about the graph transformations, we can conclude that we can obtain graph of g by starting with the graph of f and first stretching it vertically by a factor of 2 and then shifting it down 3 units. Since $f(t) = \sin(t)$ has amplitude of 1 unit, if we stretch it vertically by a factor of 2, then we'll double the amplitude, so we should expect that the amplitude of g to be 2 units. Also, since $f(t) = \sin(t)$ has midline $y = 0$, when we shift it down 3 units, the resulting midline for the function g will be $y = -3$.

period: 2π units
midline: $y = -3$
amplitude: 2 units
horizontal shift: 0 units

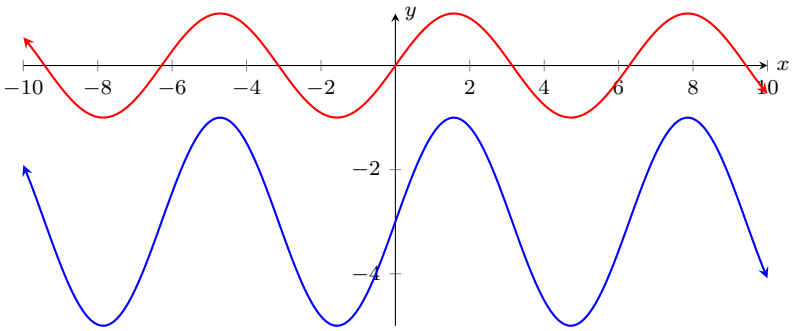

Figure 3: The graph of $f(t) = \sin(t)$ and $g(t) = 2\sin(t) - 3$

Solution 15 (Exercise 2)**.** Let's start by finding the values of A, ω, k. To do this, we need to find the midline, amplitude, and period.

The midline is $y = -3$, which dictates what the value of k is.
The amplitude is the distance from the maximum/minimum to the midline. The amplitude is 4 units.
The period is the distance from each maximum/minimum. $6.5 - 0.5 = 6$ units. The period doesn't actually appear as a number in the algebraic rule for the function. But the period is connected with the value that we use of ω. ω causes a horizontal stretch which gives us the desired period.

$$6 = 2\pi \times \frac{1}{\omega} = \frac{2\pi}{6} = \boxed{\frac{\pi}{3}}$$

. When we're trying to find a formula that involves sin, we want to think in terms of sin waves, which start at the midline and come up out of the midline. So, we'll want to find a place in this function where the function is crossing through the midline and coming up out of the midline. That would be at the point: $(0, -1)$. So, for the sin function, $h = -1$. For cos, it's the opposite. Cos starts at the maximum, and goes down to the midline. That would be at the point: $(0, 0.5)$. So, for the cos function, $h = 0.5$.

$$g(t) = 4\sin\left(\frac{\pi}{3}(t+1)\right) - 3 \qquad g(t) = 4\cos\left(\frac{\pi}{3}\left(t - \frac{1}{2}\right)\right) - 3$$

.